

Vantage Prep

SAT Math Foundations — Unit 1 Curriculum

Advanced Linear Algebra: Heart of Algebra & Linear Functions

Program	Detail
Instructors	Agastya Kondamuri / Aditya Garg
Grades	6th – 8th (Bret Harte Middle School and equivalent)
Length	1 week • 4 sessions — Mon / Wed / Thu / Fri, week of September 28, 2026. All sessions 1 hour, 7:00–8:00 PM, including Thursday Office Hours
Cohort Focus	This first cohort is 100% Advanced Linear Algebra — no quadratics — to go deep on the SAT's single most-tested skill area

1. Why Advanced Linear Algebra First

Every SAT Math test includes an Algebra domain built entirely from five official skill categories — and all five are linear-algebra topics. This isn't “extra” math on top of what Bret Harte students already do: it's the SAT's version of regular Pre-Algebra and Algebra 1 coursework, resequenced and drilled with SAT-style questions.

Official College Board skill (SAT Algebra domain)	What it means in plain language
Linear equations in one variable	Solving for x : one-step, two-step, and multi-step equations
Linear equations in two variables	Equations like $2x + 3y = 12$; intercepts, graphing
Linear functions	Slope, rate of change, $y = mx + b$, interpreting real-world meaning
Systems of two linear equations in two variables	Solving two equations at once (substitution, elimination, graphing)
Linear inequalities in one or two variables	Solving/graphing inequalities, “at least,” “at most,” constraints

Source: College Board SAT Suite, “Algebra” domain overview (satsuite.collegeboard.org).

2. How the Concepts Are Split Across the 4 Sessions

The program now runs as one intensive week: Monday, Wednesday, and Friday are full hours of real instruction, Thursday is a dedicated Office Hours hour for 1:1 support, and the final assessment is folded into Friday rather than getting its own separate day. Because class time is tighter than the old two-week format, at-home practice between sessions carries more weight than before — every session ends with a practice set that is genuinely

load-bearing, not optional extra credit. Concepts are still sequenced foundational to hardest, so nothing is taught before its prerequisite.

#	Day & Date	Time	Format	New concepts taught
1	Mon, Sept 28	7:00–8:00 PM	60 min teaching	Translating words → equations; one/two-step equations; multi-step equations (distributing, variables on both sides); linear inequalities in one variable
2	Wed, Sept 30	7:00–8:00 PM	~35 min teach + ~25 min practice	Coordinate plane & intercepts; slope (rate of change); slope-intercept form, writing equations of a line; interpreting linear functions in real-world context
3	Thu, Oct 1	7:00–8:00 PM	Office Hours	1:1 remediation on Sessions 1–2; optional small-group preview of systems of equations for students ready to move faster
4	Fri, Oct 2	7:00–8:00 PM	Teach + Word Problems + Final Test	Systems of two linear equations (substitution, elimination, special cases); guided word-problem power round; cumulative final assessment

Why this order: equations and inequalities (Session 1) are the algebraic “grammar” everything else depends on. Session 2 introduces graphing that same equation, plus what slope and intercepts mean in a real-world story — including function notation, since the Digital SAT phrases these questions with $f(x)$. Thursday’s office hour exists so no student walks into Friday shaky on Sessions 1–2; it is support time first, with a light systems preview available for students who are ready to move faster. Systems of equations (Friday) are still taught last, since they require fluency in everything before them, and Friday closes the loop with word problems and the cumulative assessment in the same session, since there is no longer a separate test day.

3. Day-by-Day Lesson Plans

Session 1 — Monday, Sept 28 • “Equation & Inequality Bootcamp” (7:00–8:00 PM, all teaching)

Objective: by the end of today, students can translate a word phrase into an equation and solve any one-variable linear equation or inequality, including ones with distribution and variables on both sides.

1. Translating words into algebra (10 min)

- “sum,” “more than,” “increased by,” “total” → +
- “difference,” “less than,” “decreased by,” “fewer” → −
- “product,” “times,” “of” → ×
- “quotient,” “per,” “divided by” → ÷

- “is,” “was,” “results in” $\rightarrow =$

"Seven less than three times a number is 20." $\rightarrow 3x - 7 = 20 \rightarrow 3x = 27 \rightarrow x = 9$

Watch for: “seven less than three times a number” is $3x - 7$, not $7 - 3x$ — the order flips. This is the single most common SAT wording trap.

2. One- and two-step equations (15 min)

$$x + 15 = -8 \rightarrow x = -23$$

$$4x - 9 = 23 \rightarrow 4x = 32 \rightarrow x = 8$$

$$x/3 + 5 = 2 \rightarrow x/3 = -3 \rightarrow x = -9$$

3. Multi-step equations — distributing & variables on both sides (20 min)

$$3(2x - 4) = 5x + 10 \rightarrow 6x - 12 = 5x + 10 \rightarrow x = 22$$

$$\text{Check: } 3(2 \cdot 22 - 4) = 120; \quad 5(22) + 10 = 120 \quad \checkmark$$

$$(2/3)x + 4 = (1/3)x + 10 \rightarrow (1/3)x = 6 \rightarrow x = 18$$

4. Linear inequalities in one variable (15 min)

$$-2x + 5 > 11 \rightarrow -2x > 6 \rightarrow x < -3 \quad (\div \text{ by a negative flips the sign — the \#1 SAT inequality trap})$$

$$3x - 4 \leq 8 \rightarrow x \leq 4$$

Graphing: open circle = $<$ or $>$; closed circle = $\leq$ or $\geq$.

Home practice (due before Session 2, Wednesday): 8 problems — 2 translation, 3 multi-step equations, 3 inequalities. Include the answer key so parents can check alongside their child. Since Wednesday's session moves fast, this set isn't optional review — it's what makes Wednesday possible in the time available.

Session 2 — Wednesday, Sept 30 • “Graphing, Slope & Linear Functions” (7:00–8:00 PM • ~35 min teach + ~25 min practice)

Objective: by the end of today, students can find intercepts, compute slope from two points, write a line's equation in slope-intercept form, and interpret a linear function in a real-world context.

5. The coordinate plane & intercepts (8 min)

$$2x + 3y = 12. \quad \text{Set } y=0 \rightarrow x\text{-intercept } (6, 0). \quad \text{Set } x=0 \rightarrow y\text{-intercept } (0, 4).$$

6. Slope — the rate of change (9 min)

$$m = (y_2 - y_1) / (x_2 - x_1). \quad \text{Points } (1,2) \text{ and } (4,11) \rightarrow m = 9/3 = 3$$

Real-world framing (this is how the SAT actually asks it): “A plant is 5 cm tall and grows 3 cm per week” $\rightarrow$ slope = 3, starting value = 5.

7. Slope-intercept form $y = mx + b$ (9 min)

Line through (2, 5) with slope -3 : $5 = -3(2) + b \rightarrow b = 11 \rightarrow y = -3x + 11$

8. Interpreting linear functions in real-world context (9 min)

A tutoring company charges $f(x) = 25x + 40$, where x = hours.
 40 = flat starting fee (y-intercept); 25 = per-hour rate (slope).
 $f(3) = 25(3) + 40 = 115$

Use function notation ($f(x)$) here even though middle school often uses y — students need to recognize both; this is exactly how the Digital SAT phrases these questions.

Practice (~25 min, individually corrected): 10 mixed problems spanning Session 1 + today's material (spiral review built in from day one), plus 2 real SAT-released questions.

Home practice (due before Thursday's office hours): 6 problems on slope/intercepts/function interpretation + 2 review problems from Session 1, to keep equations fresh going into Thursday and Friday.

Session 3 — Thursday, Oct 1 • Office Hours (7:00–8:00 PM)

Primary purpose — remediation

1:1 (or small-group) time to re-teach whatever Sessions 1–2 showed was shaky. If a majority of the group is stuck on the same concept (a common candidate: the “flip the sign” inequality rule, or reading slope from a real-world scenario), spend the first 10–15 minutes re-teaching it to everyone before splitting into individual correction. This is the moment to catch any student falling behind before Friday introduces systems of equations and the final assessment.

Secondary purpose — optional systems preview

For students who are solid on Sessions 1–2, use remaining time to informally preview substitution and elimination (the two methods taught fully on Friday). This is optional and low-stakes — no student is behind if they only use Thursday for remediation — but it means Friday's systems lesson can move faster for students who've already seen the idea once.

No new homework assigned tonight — the goal is to walk into Friday caught up, not further behind on a new set.

Session 4 — Friday, Oct 2 • Systems, Word Problems & Final Assessment (7:00–8:00 PM)

This is the longest, highest-stakes session of the program — it carries the hardest new content, the word-problem application, and the cumulative test all in one hour.

Warm-up (5 min)

Rapid-fire recap of the two trap questions from Sessions 1–2 — the “seven less than” wording trap and the “flip the sign” inequality trap — worked on the board, not re-taught from scratch.

9. Systems of two linear equations in two variables (25 min)

- Substitution: $y = 2x+1$ and $3x+y=16 \rightarrow 3x+(2x+1)=16 \rightarrow x=3, y=7 \rightarrow (3, 7)$
- Elimination: $2x+3y=12$ and $4x-3y=6 \rightarrow \text{add} \rightarrow 6x=18 \rightarrow x=3 \rightarrow y=2 \rightarrow (3, 2)$
- Special case — no solution: same slope, different intercept (parallel). E.g., $y=2x+3$ and $y=2x-1$
- Special case — infinite solutions: same slope, same intercept (same line). E.g., $y=2x+3$ and $2y=4x+6$

Word Problem Power Round (10 min)

See Section 4 below. Live-model Problem 1 (Rate/Distance/Time) and Problem 2 (Break-Even) on the board, showing how the translation skills from Session 1 and the equation/systems tools from this week combine on a real multi-step problem. The remaining Word Problem Workshop problems (Mixture, Age, Budget Constraint, Linear Function in Context) plus the take-home extension set are assigned as the program's cumulative take-home packet, sent home with a full answer key so parents can work through them with their child after the program ends.

Final Assessment (20 min)

The cumulative, real-College-Board-style assessment referenced in the program's money-back guarantee, and the last scheduled activity of the program. Every concept from Sessions 1, 2, and today is fair game, weighted toward today's material (systems of equations) and Session 2 (functions in context) since those are the newest and hardest. Mix of multiple-choice and grid-in style, matching the Digital SAT format. Score emailed to parents that day.

4. Word Problem Workshop (Friday, Word Problem Power Round + Take-Home Packet)

These six problems are the “hard but fair” tier you asked for — each requires real translation work and at least two steps, but every one is solvable with only the tools taught in Sessions 1–4. Problems 1–3 are Core (every student should finish these); 4–6 are Challenge (stretch problems for students who finish early or want extra practice at home). Problems 1 and 2 are modeled live in Friday's Word Problem Power Round; the rest go home in the cumulative take-home packet.

Strategy taught first (5 min): (1) circle what you're solving for and assign it a variable, (2) find the “is/equals” moment in the sentence, (3) translate piece by piece using the Session 1 keyword bank, (4) solve, (5) check your answer against the original story, not just the equation.

Core

1. Rate / Distance / Time

Maya leaves her house and bikes toward the library at 12 mph. At the same moment, her brother Leo leaves the library and bikes toward Maya's house at 8 mph. The library is 15 miles from Maya's house. How many minutes after they start will they meet?

Combined closing speed = $12+8 = 20$ mph. $20t = 15 \rightarrow t = 0.75$ hr = 45 minutes

2. Break-Even (linear equations)

A school club is choosing between two fundraiser vendors. Vendor A charges a \$50 setup fee plus \$3 per T-shirt. Vendor B charges a \$20 setup fee plus \$5 per T-shirt. For how many T-shirts do the two vendors cost the same, and what is that cost?

$$50 + 3x = 20 + 5x \rightarrow 30 = 2x \rightarrow x = 15 \text{ shirts; cost} = 50 + 3(15) = \$95$$

3. Mixture

A coffee shop wants to make 20 pounds of a blend selling for \$9/lb by mixing a premium bean (\$12/lb) with a standard bean (\$7/lb). How many pounds of the premium bean are needed?

$$12x + 7(20-x) = 9(20) \rightarrow 12x + 140 - 7x = 180 \rightarrow 5x = 40 \rightarrow x = 8 \text{ lbs premium (12 lbs standard)}$$

Challenge

4. Age Problem

Right now, a father is 3 times as old as his daughter. In 10 years, he will be twice as old as she will be then. How old is the daughter now?

$$3d + 10 = 2(d+10) \rightarrow 3d + 10 = 2d + 20 \rightarrow d = 10 \text{ (father is 30). Check: } 40 = 2(20) \checkmark$$

5. Budget Constraint (inequality)

Jordan has \$60 for movie tickets and popcorn. Each ticket costs \$9.50, and a large popcorn costs \$6. If Jordan buys 4 tickets, what is the maximum number of popcorns he can buy without going over budget?

$$4(9.50) = 38. \text{ Remaining} = 60 - 38 = 22. \quad 6p \leq 22 \rightarrow p \leq 3.67 \rightarrow 3 \text{ popcorns (round down)}$$

6. Linear Function in Context (Digital SAT style)

A candle is 20 cm tall when lit and burns at a constant rate. After burning for 5 hours, it is 12.5 cm tall. Write a function $h(t)$ for its height after t hours, and find how many hours until it burns out completely.

$$\text{slope} = (12.5 - 20)/5 = -1.5 \text{ cm/hr. } h(t) = -1.5t + 20. \quad h(t) = 0 \rightarrow t = 40/3 \approx 13.3 \text{ hr} = 13 \text{ hr } 20 \text{ min}$$

Take-home extension: 3 additional word problems (one of each Core type, new numbers), included in the take-home packet for students who want more reps after the program ends.

5. Parent-Facing At-a-Glance Schedule

Share this table directly with parents — it answers “what is my child learning today?” at a glance, per the program’s transparency commitment.

Day	Date	Time	Format	What your child is learning
Monday	Sept 28	7:00–8:00 PM	Teaching (60 min)	Turning word problems into equations; solving equations and

Day	Date	Time	Format	What your child is learning
				inequalities for x
Wednesday	Sept 30	7:00–8:00 PM	Teach (~35) + Practice (~25)	Graphing lines, slope as “rate of change,” writing a line’s equation, and what slope/intercept mean in real situations
Thursday	Oct 1	7:00–8:00 PM	Office Hours	1:1 help on anything missed Monday or Wednesday, plus an optional preview of solving two equations at once (systems)
Friday	Oct 2	7:00–8:00 PM	Teach + Word Problems + Final Test	Systems of equations, a guided real-world word problem, then the cumulative final assessment (counts toward the 70%+ guarantee) — last session of the program

Zoom/location details to be confirmed once finalized — this table is designed to be dropped into an enrollment email as-is.

6. Materials & At-Home Support

Materials needed each session

Pencil, notebook or graph paper (for Sessions 2 and 4), a basic or scientific calculator (not required, but allowed per the placement exam’s policy), device for Zoom.

How parents can help between sessions

You don’t need to know the math — just ask your child to explain their homework answer out loud before checking it against the answer key. If they can explain why an equation is set up the way it is, they’ve actually learned the concept, not just matched a pattern. This matters more than ever in the one-week format — with only three teaching sessions, the practice sets between them are doing real work, not just reinforcing what was already mastered in class.

Difficulty calibration note

Every practice and homework set is built in three informal tiers — Warm-up (confirms the basic mechanic), Core (SAT-typical difficulty), and Challenge (the word-problem tier above). No student sees only Warm-up or only Challenge; everyone works through all three so pacing stays consistent with the rest of the cohort while still stretching stronger students.